# DELEGATION OF FINANCIAL AUTHORITY & APPROVAL MATRIX -- DRAFT

> **Compliance Status Engine Guide**
> *Function: Programmatic Evaluation of Financial Authority Limits & Multi-Stage Compliance Workflows*
> This document translates the automated financial controls engine into sequential business rules. The engine evaluates purchase orders through multi-phase quality gates, executes AI compliance audits, routes structured email dispatches based on permissions, and continuously scans inbox replies to update the database state in real time.

## 1. Purpose and Scope

This document establishes the formal framework, financial thresholds, and authorization parameters for the expenditure of funds within the Cathkin Estates Homeowners Association. It applies to all elected Directors, Portfolio Committee Members, the Finance Committee, the Estate Manager, and the appointed Managing Agent (GEMS).

## 2. Schedule of Financial Authorities

The authorization parameters, review structures, and compliance requirements governing operational and capital outlays are structured via the system's `approval_matrix` table:

Edit Authority Matrix

| Matrix Category | Description | Min Value (ZAR) | Max Value (ZAR) | Reviewer Roles | Quotes Required | Approver Roles | Applicable Controls | Compliance Audit Requirement** |
|---|---|---|---|---|---|---|---|---|
| **Low Value Operational** | Budgeted expenditure only | R0.00 | R10,000.00 | None; Does not need to be logged in the system | 0 | Estate Manager | None | Month-end general ledger reconciliation against budgeted line items. |
| **Mid-Value Operational** | Budgeted expenditure only | R10,001.00 | R50,000.00 | Finance Committee | 2 | Chair* + Board member | Purchase Order | N/A |
| **High-Value Operational** | Budgeted expenditure only | R50,001.00 | Unlimited | Finance Committee | 3 | Chair* + 2 Board members | Purchase Order,Detailed Comparison | Recorded Board Minutes. |

| Matrix Category | Description | Min Value (ZAR) | Max Value (ZAR) | Reviewer Roles | Quotes Required | Approver Roles | Applicable Controls | Compliance Audit Requirement** |
|---|---|---|---|---|---|---|---|---|
| **Extraordinary / Unbudgeted** | Non-Emergency Maintenance / Procurement | R0.00 | R30,000.00 | Finance Committee | 3 | Full Board of Directors | Purchase Order | Written motivation presented at a formal Board meeting. |
| **Extraordinary / Unbudgeted >R30K** | Non-Emergency Maintenance / Procurement | R30,001.00 | Unlimited | Finance Committee | 3 | Full Board of Directors + Homewowner approval | Purchase Order,Detailed Comparison | Formal AGM ratification or digital homeowner vote record attached to the transaction file. |
| **Emergency Expenditure / Unbudgeted** | Critical risk to security, safety, or infrastructure | R0.00 | R30,000.00 | Finance Committee | 2 | Chair* + 1 Board member | Purchase Order | Immediate entry into the Unbudgeted Log; ratification at next Board meeting. Email notification to Homeowners |
| **Service Provider Contracts** | New Appointments or SLA Renewals | R0.00 | Unlimited | Finance Committee | 3 | Full Board of Directors | Detailed Comparison | Executed formal contract with dual signatures from authorised Directors uploaded to the transaction record. |
| **Low-Value Capital Expenditure** | Reserve Fund Allocations | R0.00 | R10,000.00 | Finance Committee | 0 | Estate Manager | None | Alignment with first year of statutory 10-Year plan. |
| **Mid-Value Capital Expenditure** | Reserve Fund Allocations | R10,001.00 | R50,000.00 | Finance Committee | 2 | Chair* + 2 Board members | Purchase Order | Alignment with first year of statutory 10-Year plan. |
| **High-Value Capital Expenditure** | Reserve Fund Allocations - Major Projects | R50,001.00 | Unlimited | Finance Committee | 3 | Full Board of Directors | Purchase Order,Detailed Comparison | Alignment with first year of statutory 10-Year plan. |

- *Chair Approval: If the Chair does not respond within 6 hours, an automated follow-up notification will be dispatched to all Board members.
- **Baseline Requirement for All Tiers, excluding Low Value Operational: Digital (email) sign-off captured automatically via the system workflow per the approval routing rules defined in this Matrix.

- All requests for approval are automatically copied to all Board members, the Finance Committee, the Estate Manager, and Administration.

## 3. Core Operational Regulations

### 3.1 The Anti-Splitting Directive

Large projects, single procurement orders, or related service invoices may not be split into multiple smaller purchase orders or partial payments to circumvent the threshold limits defined in Section 2. Threshold compliance is determined by the total cumulative cost of the required work or asset procurement.

### 3.2 Procurement Sourcing Thresholds

- **Transactions under R10,000:** A single written quote from an approved, accredited estate vendor.
- **Transactions from R10,001 to R50,000:** A minimum of two independent written quotes must be evaluated.
- **Transactions over R50,000 / Capital Tenders:** A minimum of three independent written quotes accompanied by a detailed comparison (system-generated) must be submitted to the Finance Committee.

## 4. Automated Workflow & Outbound Email Dispatch Engine

When a Purchase Order is submitted, `2_send_approvals_postgres.py` evaluates the PO status and executes automated email dispatch based on the `approval_permission` designated in the `approvers` database table:

### 4.1 Multi-Stage Dispatch & Recipient Routing

- **Stage 1: Submit for Finance Review**
  - **Primary Recipients ("To"):** All users where `approval_permission = 'Finance Review'`.
  - **Copied Recipients ("CC"):** All users where `approval_permission = 'Estate Manager'`.
  - **Required Call-To-Action Text:**

    ```
    Please review the purchase order details and reply directly to this message typing either
    "RECOMMEND FOR APPROVAL" or "RECOMMEND FOR REJECTION".
    ```

  - **Post-Dispatch Status:** Updates `submission_status` in `po_log` to `Finance Review`.
- **Stage 2: Submit for Approval**
  - **Primary Recipients ("To"):** All users where `approval_permission = 'Approval Authority'`.
  - **Copied Recipients ("CC"):** All users where `approval_permission = 'Finance Review'` AND `approval_permission = 'Estate Manager'`.
  - **Required Call-To-Action Text:**

    ```
    Please review the purchase order details and reply directly to this message typing either
    "APPROVED" or "REJECTED".
    ```

  - **Post-Dispatch Status:** Updates `submission_status` in `po_log` to `Sent`.

### 4.2 AI Recommendation Persistence

To optimize execution speed and prevent re-running costly LLM analyses, the system checks whether an `ai_recommendation_summary` already exists in the database:

- **Initial Analysis:** When sent for `Submit for Finance Review`, the Gemini AI Engine evaluates quotes, vendor OSINT records, and GL budget run-rates, storing the analysis in `po_log.ai_recommendation_summary`.
- **Summary Reuse:** When the item transitions to `Submit for Approval`, the engine detects the existing summary and automatically reuses the exact AI audit output in the approval email body without re-triggering the Gemini API.

## 5. Automated Reply Processing Engine

The background process `3_process_replies_postgres.py` continuously monitors the incoming IMAP email inbox for responses containing bracketed PO numbers (e.g., `[PO-2026-001]`) and updates `po_log` accordingly:

| Inbound Response Keyword | Current PO Status | New Database Status | Control Log Action recorded |
|---|---|---|---|
| `RECOMMEND FOR APPROVAL` / `RECOMMEND APPROVE` | Finance Review | **Finance Recommended** | Finance Review Recommendation |
| `RECOMMEND FOR REJECTION` / `RECOMMEND REJECT` | Finance Review | **Finance Rejected** | Finance Review Rejection |
| `APPROVED` / `APPROVE` | Sent / Any Active State | **Approved** | Inbound Approval |
| `REJECTED` / `REJECT` | Sent / Any Active State | **Rejected** | Inbound Rejection |

## 6. Executive Safeguards & Audit Control

- **Unalterable Audit Trails:** Every outbound email dispatch and inbound reply is logged chronologically to the PostgreSQL `workflow_control_log` table and localized audit log file.
- **Context-Aware Parsing:** The reply processor strips out historical email threads and splitters (e.g., `-----Original Message-----`) to evaluate only top-level replies from validated approvers.
- **Fiscal Runway Monitoring:** Post-purchase balances are checked against the estate's 2-Month Run-rate Buffer Pool during AI synthesis to flag structural deficit risks before approval sign-offs occur.